

15.2 Power Series

The student should pay close attention to the material because we shall show how power series play an important role in complex analysis. Indeed, they are the most important series in complex analysis because their sums are analytic functions (Theorem 5, Sec. 15.3), and every analytic function can be represented by power series (Theorem 1, Sec. 15.4).

A **power series in powers of $z - z_0$** is a series of the form

$$(1) \quad \sum_{n=0}^{\infty} a_n(z - z_0)^n = a_0 + a_1(z - z_0) + a_2(z - z_0)^2 + \cdots$$

where z is a complex variable, $a_0, a_1, \dots$ are complex (or real) constants, called the **coefficients** of the series, and z_0 is a complex (or real) constant, called the **center** of the series. This generalizes real power series of calculus.

If $z_0 = 0$, we obtain as a particular case a *power series in powers of z* :

$$(2) \quad \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots$$

Convergence Behavior of Power Series

Power series have variable terms (functions of z), but *if we fix z , then all the concepts for series with constant terms in the last section apply*. Usually a series with variable terms will converge for some z and diverge for others. For a power series the situation is simple. The series (1) may converge in a disk with center z_0 or in the whole z -plane or only at z_0 . We illustrate this with typical examples and then prove it.

EXAMPLE 1 Convergence in a Disk. Geometric Series

The *geometric series*

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \cdots$$

converges absolutely if $|z| < 1$ and diverges if $|z| \geq 1$ (see Theorem 6 in Sec. 15.1). ■

EXAMPLE 2 Convergence for Every z

The power series (which will be the Maclaurin series of e^z in Sec. 15.4)

$$\sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \cdots$$

is absolutely convergent for every z . In fact, by the ratio test, for any fixed z ,

$$\left| \frac{z^{n+1}/(n+1)!}{z^n/n!} \right| = \frac{|z|}{n+1} \rightarrow 0 \quad \text{as} \quad n \rightarrow \infty. \quad \blacksquare$$

EXAMPLE 3 Convergence Only at the Center. (Useless Series)

The following power series converges only at $z = 0$, but diverges for every $z \neq 0$, as we shall show.

$$\sum_{n=0}^{\infty} n! z^n = 1 + z + 2z^2 + 6z^3 + \cdots$$

In fact, from the ratio test we have

$$\left| \frac{(n+1)! z^{n+1}}{n! z^n} \right| = (n+1)|z| \rightarrow \infty \quad \text{as} \quad n \rightarrow \infty \quad (z \text{ fixed and } \neq 0). \quad \blacksquare$$

THEOREM 1

Convergence of a Power Series

- (a) Every power series (1) converges at the center z_0 .
- (b) If (1) converges at a point $z = z_1 \neq z_0$, it converges absolutely for every z closer to z_0 than z_1 , that is, $|z - z_0| < |z_1 - z_0|$. See Fig. 365.
- (c) If (1) diverges at $z = z_2$, it diverges for every z farther away from z_0 than z_2 . See Fig. 365.

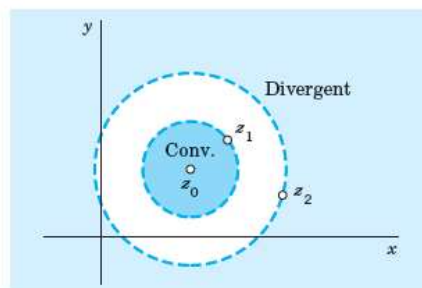


Fig. 365. Theorem 1

Radius of Convergence of a Power Series

Convergence for every z (the nicest case, Example 2) or for no $z \neq z_0$ (the useless case, Example 3) needs no further discussion, and we put these cases aside for a moment. We consider the *smallest* circle with center z_0 that includes all the points at which a given power series (1) converges. Let R denote its radius. The circle

$$|z - z_0| = R \quad (\text{Fig. 366})$$

is called the **circle of convergence** and its radius R the **radius of convergence** of (1). Theorem 1 then implies convergence everywhere within that circle, that is, for all z for which

$$(4) \quad |z - z_0| < R$$

(the open disk with center z_0 and radius R). Also, since R is as *small* as possible, the series (1) diverges for all z for which

$$(5) \quad |z - z_0| > R.$$

No general statements can be made about the convergence of a power series (1) *on the circle of convergence* itself. The series (1) may converge at some or all or none of the points. Details will not be important to us. Hence a simple example may just give us the idea.

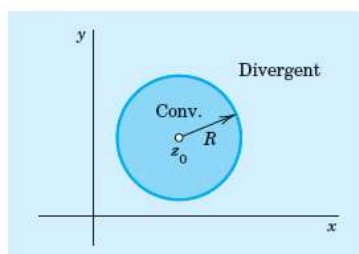


Fig. 366. Circle of convergence

EXAMPLE 4 Behavior on the Circle of Convergence

On the circle of convergence (radius $R = 1$ in all three series),

$\sum z^n/n^2$ converges everywhere since $\sum 1/n^2$ converges,

$\sum z^n/n$ converges at -1 (by Leibniz's test) but diverges at 1 ,

$\sum z^n$ diverges everywhere.

* p series or the hypergeometric Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

if $p \leq 1$ it diverges

Notations $R = \infty$ and $R = 0$. To incorporate these two excluded cases in the present notation, we write

$R = \infty$ if the series (1) converges for all z (as in Example 2),

$R = 0$ if (1) converges only at the center $z = z_0$ (as in Example 3).

These are convenient notations, but nothing else.

but if $p > 1$ it converges

Real Power Series. In this case in which powers, coefficients, and center are real, formula (4) gives the **convergence interval** $|x - x_0| < R$ of length $2R$ on the real line.

Determination of the Radius of Convergence from the Coefficients. For this important practical task we can use

THEOREM 2

Radius of Convergence R

Suppose that the sequence $|a_{n+1}/a_n|$, $n = 1, 2, \dots$, converges with limit L^* . If $L^* = 0$, then $R = \infty$; that is, the power series (1) converges for all z . If $L^* \neq 0$ (hence $L^* > 0$), then

$$(6) \quad R = \frac{1}{L^*} = \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| \quad (\text{Cauchy-Hadamard formula}^1).$$

If $|a_{n+1}/a_n| \rightarrow \infty$, then $R = 0$ (convergence only at the center z_0).

PROOF For (1) the ratio of the terms in the ratio test (Sec. 15.1) is

$$\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| = \left| \frac{a_{n+1}}{a_n} \right| |z - z_0|. \quad \text{The limit is} \quad L = L^* |z - z_0|.$$

Let $L^* \neq 0$, thus $L^* > 0$. We have convergence if $L = L^* |z - z_0| < 1$, thus $|z - z_0| < 1/L^*$, and divergence if $|z - z_0| > 1/L^*$. By (4) and (5) this shows that $1/L^*$ is the convergence radius and proves (6).

If $L^* = 0$, then $L = 0$ for every z , which gives convergence for all z by the ratio test. If $|a_{n+1}/a_n| \rightarrow \infty$, then $|a_{n+1}/a_n| |z - z_0| > 1$ for any $z \neq z_0$ and all sufficiently large n . This implies divergence for all $z \neq z_0$ by the ratio test (Theorem 7, Sec. 15.1). ■

Formula (6) will not help if L^* does not exist, but extensions of Theorem 2 are still possible, as we discuss in Example 6 below.

EXAMPLE 5 Radius of Convergence

By (6) the radius of convergence of the power series $\sum_{n=0}^{\infty} \frac{(2n)!}{(n!)^2} (z - 3i)^n$ is

$$R = \lim_{n \rightarrow \infty} \left[\frac{(2n)!}{(n!)^2} \bigg/ \frac{(2n+2)!}{((n+1)!)^2} \right] = \lim_{n \rightarrow \infty} \left[\frac{(2n)!}{(2n+2)!} \cdot \frac{((n+1)!)^2}{(n!)^2} \right] = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(2n+2)(2n+1)} = \frac{1}{4}.$$

The series converges in the open disk $|z - 3i| < \frac{1}{4}$ of radius $\frac{1}{4}$ and center $3i$. ■

Example #06 Extension of Theorem 2.

Find the radius of convergence R of the Power series.

$$\sum_{n=0}^{\infty} \left[1 + (-1)^n + \frac{1}{2^n} \right] z^n = 3 + \frac{1}{2}z + \left(2 + \frac{1}{4}\right)z^2 + \frac{1}{8}z^3 + \left(2 + \frac{1}{16}\right)z^4 + \dots$$

Sol:-

$$\sum_{n=0}^{\infty} \left[1 + (-1)^n + \frac{1}{2^n} \right] z^n = \left(2 + \frac{1}{2^0}\right) + \frac{1}{2}z + \left(2 + \frac{1}{4}\right)z^2 + \frac{1}{8}z^3 + \left(2 + \frac{1}{16}\right)z^4 + \dots$$

The sequences of the ratios $\frac{1}{6}$, $2\left(2 + \frac{1}{4}\right)$,

$\frac{1}{8}\left(2 + \frac{1}{4}\right)$, $\dots$ does not converge, so that

Theorem (2) is of no help. It can be shown that

$$R = \frac{1}{L}, \quad L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$$

Consider for odd n .

This still does not help here. Since

$(\sqrt[n]{|a_n|})$ does not converge because

$$\sqrt[n]{|a_n|} = \sqrt[n]{1/2^n} = 1/2 \quad \text{for odd } n.$$

whereas for even n we have

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sqrt[n]{2 + \frac{1}{2^n}} = \lim_{n \rightarrow \infty} \left(2 + \frac{1}{2^n}\right)^{\frac{1}{n}} = 1 \text{ as}$$

So that

$\sqrt[n]{|a_n|}$ has the two limit points

$\frac{1}{2}$ and 1 . It can further be shown that

$R = \frac{1}{\tilde{L}}$ $\tilde{L}$ is the greatest point of the sequence $\{\sqrt[n]{|a_n|}\}$.

Here $\tilde{L} = 1$ so that $R = 1$

The series converges for $|z| < 1$

Summary:- Power series converges in an open circular disk or some even for every z (or some only at the center, but they are use less); for the radius of convergence. See Ex # 6.

Except for the use less ones, power series have sums that are analytic functions.

Recommended Exercise 15.2.

15.3 Functions Given by Power Series

Here, our main goal is to show that power series represent analytic functions. This fact (Theorem 5) and the fact that power series behave nicely under addition, multiplication, differentiation, and integration accounts for their usefulness.

To simplify the formulas in this section, we take $z_0 = 0$ and write

$$(1) \quad \sum_{n=0}^{\infty} a_n z^n.$$

There is no loss of generality because a series in powers of $\hat{z} - z_0$ with any z_0 can always be reduced to the form (1) if we set $\hat{z} - z_0 = z$.

Terminology and Notation. If any given power series (1) has a nonzero radius of convergence R (thus $R > 0$), its sum is a function of z , say $f(z)$. Then we write

$$(2) \quad f(z) = \sum_{n=0}^{\infty} a_n z^n = a_0 + a_1 z + a_2 z^2 + \cdots \quad (|z| < R).$$

We say that $f(z)$ is **represented** by the power series or that it is **developed** in the power series. For instance, the geometric series *represents* the function $f(z) = 1/(1 - z)$ in the interior of the unit circle $|z| = 1$. (See Theorem 6 in Sec. 15.1.)

Uniqueness of a Power Series Representation. This is our next goal. It means that a function $f(z)$ cannot be represented by two different power series with the same center. We claim that if $f(z)$ can at all be developed in a power series with center z_0 , the development is unique. This important fact is frequently used in complex analysis (as well as in calculus). We shall prove it in Theorem 2. The proof will follow from

THEOREM 1

Continuity of the Sum of a Power Series

If a function $f(z)$ can be represented by a power series (2) with radius of convergence $R > 0$, then $f(z)$ is continuous at $z = 0$.

THEOREM 2

Identity Theorem for Power Series. Uniqueness

Let the power series $a_0 + a_1 z + a_2 z^2 + \cdots$ and $b_0 + b_1 z + b_2 z^2 + \cdots$ both be convergent for $|z| < R$, where R is positive, and let them both have the same sum for all these z . Then the series are identical, that is, $a_0 = b_0, a_1 = b_1, a_2 = b_2, \dots$.

Hence if a function $f(z)$ can be represented by a power series with any center z_0 , this representation is **unique**.

Operations on Power Series

Interesting in itself, this discussion will serve as a preparation for our main goal, namely, to show that functions represented by power series are analytic.

Termwise addition or subtraction of two power series with radii of convergence R_1 and R_2 yields a power series with radius of convergence at least equal to the smaller of R_1 and R_2 . *Proof.* Add (or subtract) the partial sums s_n and s_n^* term by term and use $\lim (s_n \pm s_n^*) = \lim s_n \pm \lim s_n^*$.

Termwise multiplication of two power series

$$f(z) = \sum_{k=0}^{\infty} a_k z^k = a_0 + a_1 z + \cdots$$

and

$$g(z) = \sum_{m=0}^{\infty} b_m z^m = b_0 + b_1 z + \cdots$$

means the multiplication of each term of the first series by each term of the second series and the collection of like powers of z . This gives a power series, which is called the **Cauchy product** of the two series and is given by

$$\begin{aligned} a_0 b_0 + (a_0 b_1 + a_1 b_0)z + (a_0 b_2 + a_1 b_1 + a_2 b_0)z^2 + \cdots \\ = \sum_{n=0}^{\infty} (a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0)z^n. \end{aligned}$$

We mention without proof that this power series converges absolutely for each z within the smaller circle of convergence of the two given series and has the sum $s(z) = f(z)g(z)$. For a proof, see [D5] listed in App. 1.

Termwise differentiation and integration of power series is permissible, as we show next. We call **derived series of the power series** (1) the power series obtained from (1) by termwise differentiation, that is,

$$(3) \quad \sum_{n=1}^{\infty} n a_n z^{n-1} = a_1 + 2a_2 z + 3a_3 z^2 + \cdots.$$

THEOREM

Termwise Differentiation of a Power Series

The derived series of a power series has the same radius of convergence as the original series.

EXAMPLE 1 Application of Theorem 3

Find the radius of convergence R of the following series by applying Theorem 3.

$$\sum_{n=2}^{\infty} \binom{n}{2} z^n = z^2 + 3z^3 + 6z^4 + 10z^5 + \cdots$$

Solution. Differentiate the geometric series twice term by term and multiply the result by $z^2/2$. This yields the given series. Hence $R = 1$ by Theorem 3. ■

THEOREM 4

Termwise Integration of Power Series

The power series

$$\sum_{n=0}^{\infty} \frac{a_n}{n+1} z^{n+1} = a_0 z + \frac{a_1}{2} z^2 + \frac{a_2}{3} z^3 + \cdots$$

obtained by integrating the series $a_0 + a_1 z + a_2 z^2 + \cdots$ term by term has the same radius of convergence as the original series.

The proof is similar to that of Theorem 3.

With the help of Theorem 3, we establish the main result in this section.

Power Series Represent Analytic Functions

THEOREM 5

Analytic Functions. Their Derivatives

A power series with a nonzero radius of convergence R represents an analytic function at every point interior to its circle of convergence. The derivatives of this function are obtained by differentiating the original series term by term. All the series thus obtained have the same radius of convergence as the original series. Hence, by the first statement, each of them represents an analytic function.

Summary. The results in this section show that power series are about as nice as we could hope for: we can differentiate and integrate them term by term (Theorems 3 and 4). Theorem 5 accounts for the great importance of power series in complex analysis: the sum of such a series (with a positive radius of convergence) is an analytic function and has derivatives of all orders, which thus in turn are analytic functions. But this is only part of the story. In the next section we show that, conversely, *every* given analytic function $f(z)$ can be represented by power series, called *Taylor series* and being the complex analog of the real Taylor series of calculus.

15.4 Taylor and Maclaurin Series

The **Taylor series**³ of a function $f(z)$, the complex analog of the real Taylor series is

$$(1) \quad f(z) = \sum_{n=1}^{\infty} a_n(z - z_0)^n \quad \text{where} \quad a_n = \frac{1}{n!} f^{(n)}(z_0)$$

or, by (1), Sec. 14.4,

$$(2) \quad a_n = \frac{1}{2\pi i} \oint_C \frac{f(z^*)}{(z^* - z_0)^{n+1}} dz^*.$$

In (2) we integrate counterclockwise around a simple closed path C that contains z_0 in its interior and is such that $f(z)$ is analytic in a domain containing C and every point inside C .

A **Maclaurin series**³ is a Taylor series with center $z_0 = 0$.

The **remainder** of the Taylor series (1) after the term $a_n(z - z_0)^n$ is

$$(3) \quad R_n(z) = \frac{(z - z_0)^{n+1}}{2\pi i} \oint_C \frac{f(z^*)}{(z^* - z_0)^{n+1}(z^* - z)} dz^*$$

(proof below). Writing out the corresponding partial sum of (1), we thus have

$$(4) \quad \begin{aligned} f(z) = f(z_0) + \frac{z - z_0}{1!} f'(z_0) + \frac{(z - z_0)^2}{2!} f''(z_0) + \cdots \\ + \frac{(z - z_0)^n}{n!} f^{(n)}(z_0) + R_n(z). \end{aligned}$$

This is called **Taylor's formula with remainder**.

We see that *Taylor series are power series*. From the last section we know that power series represent analytic functions. And we now show that *every* analytic function can be represented by power series, namely, by Taylor series (with various centers). This makes Taylor series very important in complex analysis. Indeed, they are more fundamental in complex analysis than their real counterparts are in calculus.

THEOREM 1

Taylor's Theorem

Let $f(z)$ be analytic in a domain D , and let $z = z_0$ be any point in D . Then there exists precisely one Taylor series (1) with center z_0 that represents $f(z)$. This representation is valid in the largest open disk with center z_0 in which $f(z)$ is analytic. The remainders $R_n(z)$ of (1) can be represented in the form (3). The coefficients satisfy the inequality

$$(5) \quad |a_n| \leq \frac{M}{r^n}$$

where M is the maximum of $|f(z)|$ on a circle $|z - z_0| = r$ in D whose interior is also in D .

Accuracy of Approximation. We can achieve any preassigned accuracy in approximating $f(z)$ by a partial sum of (1) by choosing n large enough. This is the practical use of formula (9).

Singularity, Radius of Convergence. On the circle of convergence of (1) there is at least one **singular point** of $f(z)$, that is, a point $z = c$ at which $f(z)$ is not analytic (but such that every disk with center c contains points at which $f(z)$ is analytic). We also say that $f(z)$ is **singular** at c or **has a singularity** at c . Hence the radius of convergence R of (1) is usually equal to the distance from z_0 to the nearest singular point of $f(z)$.

(Sometimes R can be greater than that distance: $\ln z$ is singular on the negative real axis, whose distance from $z_0 = -1 + i$ is 1, but the Taylor series of $\ln z$ with center $z_0 = -1 + i$ has radius of convergence $\sqrt{2}$.)

Power Series as Taylor Series

Taylor series are power series—of course! Conversely, we have

THEOREM 2

Relation to the Previous Section

A power series with a nonzero radius of convergence is the Taylor series of its sum.

Comparison with Real Functions. One surprising property of complex analytic functions is that they have derivatives of all orders, and now we have discovered the other surprising property that they can always be represented by power series of the form (1). This is not true in general for *real functions*; there are real functions that have derivatives of all orders but cannot be represented by a power series. (Example: $f(x) = \exp(-1/x^2)$ if $x \neq 0$ and $f(0) = 0$; this function cannot be represented by a Maclaurin series in an open disk with center 0 because all its derivatives at 0 are zero.)

Important Special Taylor Series

These are as in calculus, with x replaced by complex z . Can you see why? (*Answer.* The coefficient formulas are the same.)

EXAMPLE 1 Geometric Series

Let $f(z) = 1/(1 - z)$. Then we have $f^{(n)}(z) = n!/(1 - z)^{n+1}$, $f^{(n)}(0) = n!$. Hence the Maclaurin expansion of $1/(1 - z)$ is the geometric series

$$(11) \quad \frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n = 1 + z + z^2 + \cdots \quad (|z| < 1).$$

$f(z)$ is singular at $z = 1$; this point lies on the circle of convergence. ■

EXAMPLE 2 Exponential Function

We know that the exponential function e^z (Sec. 13.5) is analytic for all z , and $(e^z)' = e^z$. Hence from (1) with $z_0 = 0$ we obtain the Maclaurin series

$$(12) \quad e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} = 1 + z + \frac{z^2}{2!} + \cdots$$

This series is also obtained if we replace x in the familiar Maclaurin series of e^x by z .

Furthermore, by setting $z = iy$ in (12) and separating the series into the real and imaginary parts (see Theorem 2, Sec. 15.1) we obtain

$$e^{iy} = \sum_{n=0}^{\infty} \frac{(iy)^n}{n!} = \sum_{k=0}^{\infty} (-1)^k \frac{y^{2k}}{(2k)!} + i \sum_{k=0}^{\infty} (-1)^k \frac{y^{2k+1}}{(2k+1)!}.$$

Since the series on the right are the familiar Maclaurin series of the real functions $\cos y$ and $\sin y$, this shows that we have rediscovered the **Euler formula**

$$(13) \quad e^{iy} = \cos y + i \sin y.$$

Indeed, one may use (12) for *defining* e^z and derive from (12) the basic properties of e^z . For instance, the differentiation formula $(e^z)' = e^z$ follows readily from (12) by termwise differentiation. ■

EXAMPLE 3 Trigonometric and Hyperbolic Functions

By substituting (12) into (1) of Sec. 13.6 we obtain

$$(14) \quad \begin{aligned} \cos z &= \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} = 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - + \cdots \\ \sin z &= \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} = z - \frac{z^3}{3!} + \frac{z^5}{5!} - + \cdots. \end{aligned}$$

When $z = x$ these are the familiar Maclaurin series of the real functions $\cos x$ and $\sin x$. Similarly, by substituting (12) into (11), Sec. 13.6, we obtain

$$(15) \quad \begin{aligned} \cosh z &= \sum_{n=0}^{\infty} \frac{z^{2n}}{(2n)!} = 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \cdots \\ \sinh z &= \sum_{n=0}^{\infty} \frac{z^{2n+1}}{(2n+1)!} = z + \frac{z^3}{3!} + \frac{z^5}{5!} + \cdots. \end{aligned}$$

EXAMPLE 4 Logarithm

From (1) it follows that

$$(16) \quad \operatorname{Ln}(1+z) = z - \frac{z^2}{2} + \frac{z^3}{3} - + \cdots \quad (|z| < 1).$$

Replacing z by $-z$ and multiplying both sides by -1 , we get

$$(17) \quad -\operatorname{Ln}(1-z) = \operatorname{Ln} \frac{1}{1-z} = z + \frac{z^2}{2} + \frac{z^3}{3} + \cdots \quad (|z| < 1).$$

By adding both series we obtain

$$(18) \quad \operatorname{Ln} \frac{1+z}{1-z} = 2 \left(z + \frac{z^3}{3} + \frac{z^5}{5} + \cdots \right) \quad (|z| < 1). \quad \blacksquare$$

Practical Methods

The following examples show ways of obtaining Taylor series more quickly than by the use of the coefficient formulas. Regardless of the method used, the result will be the same. This follows from the uniqueness (see Theorem 1).

EXAMPLE 5 Substitution

Find the Maclaurin series of $f(z) = 1/(1 + z^2)$.

Solution. By substituting $-z^2$ for z in (11) we obtain

$$(19) \quad \frac{1}{1 + z^2} = \frac{1}{1 - (-z^2)} = \sum_{n=0}^{\infty} (-z^2)^n = \sum_{n=0}^{\infty} (-1)^n z^{2n} = 1 - z^2 + z^4 - z^6 + \cdots \quad (|z| < 1). \quad \blacksquare$$

EXAMPLE 6 Integration

Find the Maclaurin series of $f(z) = \arctan z$.

Solution. We have $f'(z) = 1/(1 + z^2)$. Integrating (19) term by term and using $f(0) = 0$ we get

$$\arctan z = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} z^{2n+1} = z - \frac{z^3}{3} + \frac{z^5}{5} - \cdots \quad (|z| < 1);$$

this series represents the principal value of $w = u + iv = \arctan z$ defined as that value for which $|u| < \pi/2$. $\blacksquare$

EXAMPLE 7 Development by Using the Geometric Series

Develop $1/(c - z)$ in powers of $z - z_0$, where $c - z_0 \neq 0$.

Solution. This was done in the proof of Theorem 1, where $c = z^*$. The beginning was simple algebra and then the use of (11) with z replaced by $(z - z_0)/(c - z_0)$:

$$\begin{aligned} \frac{1}{c - z} &= \frac{1}{c - z_0 - (z - z_0)} = \frac{1}{(c - z_0) \left(1 - \frac{z - z_0}{c - z_0} \right)} = \frac{1}{c - z_0} \sum_{n=0}^{\infty} \left(\frac{z - z_0}{c - z_0} \right)^n \\ &= \frac{1}{c - z_0} \left(1 + \frac{z - z_0}{c - z_0} + \left(\frac{z - z_0}{c - z_0} \right)^2 + \cdots \right). \end{aligned}$$

This series converges for

$$\left| \frac{z - z_0}{c - z_0} \right| < 1, \quad \text{that is,} \quad |z - z_0| < |c - z_0|. \quad \blacksquare$$

Example #08

Find the Taylor series of the following function with center $z_0=1$

$$f(z) = \frac{2z^2 + 9z + 5}{z^3 + z^2 - 8z - 12}$$

Sol:-

$$f(z) = \frac{2z^2 + 9z + 5}{(z-3)(z+2)^2}$$

Consider

$$\frac{2z^2 + 9z + 5}{(z-3)(z+2)^2} = \frac{A}{z-3} + \frac{B}{z+2} + \frac{C}{(z+2)^2}$$

$$2z^2 + 9z + 5 = A(z+2)^2 + B(z+2)(z-3) + C(z-3) \rightarrow (*)$$

put $z=3$

$$18 + 27 + 5 = A(5)^2 + B(0) + C(0)$$

$$50 = 25A \Rightarrow A = 2$$

from (*)

put $z=-2$ in eq (*)

$$8 - 18 + 5 = A(0) + B(0) + C(-2-3)$$
$$-10 = C(-5)$$

By Synthetic division

$$z^3 + z^2 - 8z - 12 \quad \text{let } z=3$$

27 + 9 - 24 - 12					36
0					
3	1	1	-8	-12	the root of the above eq
		3	12	12	
	1	4	4	0	

$$z^2 + 4z + 4$$

$$(z)^2 + 2(z)(2) + (2)^2$$
$$(z+2)^2$$

From eq (★) we have.

$$2z^2 + 9z + 5 = A[z^2 + 4 + 2z] + B[z^2 - 3z + 2z - 6] + C[z - 3]$$

By comparing coefficient.

$$2z^2$$

$$2 = A + B$$

$$B = 2 - A$$

$$1B = 2 - 2 \Rightarrow B = 0$$

from eq (1) we have.

$$f(z) \frac{2z^2 + 9z + 5}{(z+2)^2(z-3)} = \frac{2}{z-3} + \frac{1}{(z+2)^2} \rightarrow (1)$$

As we know that Binomial series.

$$\frac{1}{(1+z)^m} = 1 - mz + \frac{m(m+1)}{2!} z^2 - \frac{m(m+1)(m+2)}{3!} z^3 \dots (2)$$

from eq (1) part (ii)

$$\frac{1}{(z+2)^2} = \frac{1}{[3 + (z-1)]^2} = \frac{1}{3^2 \left[1 + \left(\frac{z-1}{3}\right)\right]^2} \rightarrow (1*)$$

using eq (2)

$$\Rightarrow \frac{1}{9} \left[1 - 2\left(\frac{z-1}{3}\right) + \frac{2(2+1)}{2!} \left[\left(\frac{z-1}{3}\right)^2\right] - \frac{2(2+1)(2+2)}{3!} \left[\left(\frac{z-1}{3}\right)^3\right] \dots \right]$$

(★)

ex ① part 1

$$\frac{2}{z-3} = \frac{2}{(z-1)-2} = \frac{2}{-2\left[1-\frac{(z-1)}{2}\right]} = -\frac{1}{\left[1-\frac{(z-1)}{2}\right]}$$

As $\frac{1}{1-z} = 1 + z + z^2 + \dots$

so $-\frac{1}{\left[1-\frac{(z-1)}{2}\right]} = -\left[1 + \frac{(z-1)}{2} + \frac{(z-1)^2}{4} + \dots\right] \quad \text{---} \textcircled{**}$

Invoking $\textcircled{*}$ and $\textcircled{**}$ in ex ①

$$\Rightarrow \left[\frac{1}{9} - \frac{2}{3 \times 9} (z-1) + \frac{1}{9 \times 9} (z-1)^2 \dots \right] - 1 - \frac{(z-1)}{2} - \frac{(z-1)^2}{4} \dots$$

$$\Rightarrow \left(\frac{1}{9} - 1 \right) - (z-1) \left[\frac{2}{27} + \frac{1}{2} \right] + \left[\frac{3}{81} - \frac{1}{4} \right] (z-1)^2 \dots$$

$$\Rightarrow -\frac{8}{9} - \frac{31}{54} (z-1) - \frac{23}{108} (z-1)^2 \dots$$

We see that the series converges $\textcircled{**}$ for $|z-1| < 3$ and the second for $|z-1| < 2$

This had to be expected because $\frac{1}{(z+2)^2}$

is singular at -2 and $\frac{2}{z-3}$ at 3 and

these points have distance 3 and 2 respectively

from the center $z=1$. Hence the whole series

converges for $|z-1| < 2$.

